Lab 11 | Sharding and Horizontal Scaling

Shard key evaluation, two-shard hotspot experiment, and MongoDB resharding activity

Decision matrix

Candidate	Cardinality / distribution	Query targeting	Verdict
order_id (random UUID)	Very high; sequential IDs may hotspot	Order lookup targets; customer/date queries scatter	Good for ID lookups
customer_id (ranged)	High; heavy customers can skew	Customer equality targets; ranges target chunks	Good locality; monitor hot customers
order_date	Monotonic; new writes go to upper range	Date ranges target relevant chunks	Avoid as sole write-heavy key
customer_id (hashed)	High; spreads inserts	Customer equality targets; ranges scatter	**Recommended** for customer-history workload

Measured run | MongoDB 7.0.45 | 10,000 documents per collection

Collection	Shard counts / data size	Range query targets
hotspot_data (ts: 1)	shard1RS: 10,000; 2,230,000 bytes	1 shard (indexed range)
even_data (user_id: hashed)	shard1RS: 5,002; shard2RS: 4,998; total 2,230,000 bytes	2 shards (scatter-gather)

Resharding note

MongoDB 5.0+ can change a collection shard key with reshardCollection. Check unique-index compatibility and resource headroom, issue it through mongos, and monitor with $currentOp. Resharding clones data, builds indexes, catches up writes, then cuts over; plan for extra resources and a brief write pause. Queryable Encryption collections are unsupported; Atlas Search indexes may need rebuilding.

Docker Hub image

Repository: https://hub.docker.com/r/krish1226/lab10 | **Tag:** lab11

```
docker pull krish1226/lab10:lab11
```

Start all services with docker compose up --build -d, then run the Lab 11 exercise using docker compose --profile lab11 run --rm lab11. Full details and measured plans are in docs/lab11.md and docs/lab11-results.json.